

Week 1

1. Introduction to Economics

- **Economics is the science of decision-making.**
- **Definition (Alfred Marshall):** The study of human behavior in the ordinary business of life.
- **Managerial Economics:** The application of **economic principles** to business decision-making.

2. Why Should We Study Economics?

- Helps managers **allocate scarce resources** efficiently.
- Provides a **framework** to solve real-world business problems.
- **Three disciplines contribute to economics:**
 - **Statistics:** Econometrics, causal inference.
 - **Computer Science:** Computational social science.
 - **Economics:** A social and business science.

3. What is Managerial Economics?

- **Managerial Economics = Microeconomics + Decision Science**
- **Managers** direct resources to achieve organizational goals.
- **Key Areas Covered:**
 - Consumer behavior
 - Market structures
 - Pricing strategies
 - Government interventions
 - Platform businesses (e.g., Uber, Amazon)

4. Decision-Making in Economics

- **Key Questions:**
 1. Who made the wrong/bad decision?
 2. Did the decision-maker have enough information?
 3. Did they have the right incentives?
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5. Economic Models

- **Models reduce complexity** (like a map simplifies geography).
- **Types of Models:**
 - **Mathematical** (Optimization, Equilibrium)
 - **Statistical** (Regression, Forecasting)

Example: Pricing Decision

- **Problem:** You own an apartment complex.

- **Demand:** Maximum rent willingness (in hundred per month) from different tenants:

Name	A	B	C	D	E	F	G
Rent	70	65	60	55	40	35	30

- **Insights from Data:**
 - Higher price → Fewer tenants.
 - Lower price → More tenants but lower revenue.
 - **Optimal price must balance demand and revenue.**

6. Rational Buyers and Consumer Surplus

- **Consumers are rational** and make decisions based on:
 1. **Preferences** – Consistency in choices.
 2. **Reservation Price** – The maximum amount a consumer is willing to pay.
 3. **Consumer Surplus** – The difference between willingness to pay and actual price.

Formula for Consumer Surplus:

Consumer Surplus = Reservation Price – Actual Price

Example

- A buyer is willing to pay ₹70 for a product.
- The seller charges ₹50.
- **Consumer Surplus** = ₹70 - ₹50 = ₹20.

7. Demand and Supply

- **Demand Curve:** Downward sloping → As price **decreases**, quantity **increases**.
- **Supply Curve:** Upward sloping → As price **increases**, quantity **increases**.

Key Terms

- **Choke Price:** The price at which demand drops to **zero**.
- **Market Equilibrium:** Where demand = supply.

8. Seller's Perspective: Profit Maximization

- **Goal:** Maximize **profit**, which depends on:
 - **Revenue (Price × Quantity)**
 - **Cost (Raw materials, technology, wages, etc.)**

Formula for Profit:

Profit = Total Revenue – Total Cost

9. Market Structures

Market Type	Features	Example
Monopoly	One seller, many buyers	Google (Search Engine)
Monopsony	One buyer, many sellers	Defense contracts (Government)
Oligopoly	Few sellers, many buyers	Telecom Industry
Perfect Competition	Many sellers, many buyers	Agricultural Markets

10. Application Example: Decision-Making in Pricing

Scenario:

- You receive a **free concert ticket** for Sonu Nigam (worth ₹7500, but can only sell for ₹5000).
- You later learn that **Shreya Ghosal** is also performing the same day, and tickets cost ₹7500.
- Question:** Should you attend Sonu Nigam's concert or buy a ticket for Shreya Ghosal?

Concept Applied: Opportunity Cost

- Opportunity Cost** = Value of the best alternative foregone.
- Since you would have preferred Shreya Ghosal's concert, the **opportunity cost of attending Sonu Nigam's concert is ₹7500**, making it a suboptimal choice.

11. Case Study: Travel Cost Comparison

Problem:

- You are planning a **500 KM trip** to Delhi.
- Options:
 - Train:** ₹1000 fare.
 - Car Rental Estimate:**

Cost Component	Annual Cost (INR)
Insurance	10,000
Interest	20,000
Fuel & Oil	10,000
Maintenance	10,000
Total	50,000

Concepts Applied:

- Opportunity Cost** – What is the real cost of driving?
- Implicit & Explicit Costs** – Are hidden costs considered?

12. Economic vs. Accounting Profit

Concept	Formula	Example
Accounting Profit	Total Revenue - Accounting Costs	₹635 - ₹312 = ₹323 Crores
Economic Profit	Total Revenue - (Accounting Costs + Opportunity Costs)	₹635 - (₹312 + Opportunity Cost)

13. Lessons from Week 1

- ✓ Understand rational decision-making and consumer surplus.
- ✓ Price affects both demand and revenue—find the optimal balance.
- ✓ Opportunity cost plays a crucial role in economic decisions.
- ✓ Market structures determine pricing and competition.
- ✓ Differentiate between accounting and economic profit.

 **Week 1: Introduction to Economics**

1. Consumer Surplus

Consumer Surplus = Reservation Price – Actual Price

2. Profit Calculation

Profit = Total Revenue – Total Cost

3. Market Equilibrium

$Q_d = Q_s$

- Q_d = Quantity demanded
- Q_s = Quantity supplied

Week 2

1. Opportunity Cost and Sunk Cost

Opportunity Cost

- **Definition:** The value of the **best alternative foregone** when making a decision.
- **Key Idea:** What you **give up** when you choose one option over another.
- **Example:**
 - You invest ₹1,00,000 in stocks instead of putting it in a fixed deposit.
 - The **opportunity cost** = Interest you could have earned on the deposit.

Sunk Cost

- **Definition:** Costs that have already been **incurred and cannot be recovered**.
- **Key Idea:** **Ignore sunk costs** when making future decisions.

- **Example:**
 - You buy a ₹5000 movie ticket but don't feel like going.
 - The ₹5000 is a **sunk cost** – it should not affect your decision to go or not.

Opportunity Cost vs. Sunk Cost

Concept	Definition	Example
Opportunity Cost	Value of the best alternative forgone	Skipping work to attend a concert
Sunk Cost	Irrecoverable past expense	Non-refundable gym membership

2. Cost Concepts in Decision Making

Types of Costs

Cost Type	Definition	Example
Fixed Cost (FC)	Costs that do not change with output	Rent, Salaries
Variable Cost (VC)	Costs that change with production	Raw materials, Labor
Total Cost (TC)	Sum of FC and VC	$TC = FC + VC$
Average Cost (AC)	Cost per unit	$AC = TC / Q$
Marginal Cost (MC)	Extra cost of producing one more unit	$MC = \Delta TC / \Delta Q$

3. Short-Run vs. Long-Run Costs

Feature	Short Run	Long Run
Fixed Costs	Exists	Can be changed
Decision Making	Limited flexibility	More flexibility
Example	Factory rent	Building a new factory

4. Law of Diminishing Marginal Returns

- **Definition:** Adding more of one input (while holding others constant) **eventually decreases additional output.**
- **Example:**
 - **Factory with 1 machine** → 1 worker is efficient.
 - **Adding more workers** → Productivity increases at first.
 - **After a point,** too many workers **reduce efficiency** due to crowding.

5. Production Function and Cost Curves

- **Production Function:** Relationship between **inputs (labor, capital)** and **output**.

$$Q=f(L,K)$$

- **Q** = Output
- **L** = Labor
- **K** = Capital

Cost Curves

Curve	Definition	Formula
Average Total Cost (ATC)	Total cost per unit	$ATC=TC/Q$
Average Variable Cost (AVC)	Variable cost per unit	$AVC=VC/Q$
Marginal Cost (MC)	Additional cost of one more unit	$MC=\Delta TC/\Delta Q$

6. Economies and Diseconomies of Scale

Economies of Scale

- **Definition:** Cost per unit **decreases** as production **increases**.
- **Example:**
 - A **large factory** buys raw materials in bulk, reducing per-unit cost.

Diseconomies of Scale

- **Definition:** Cost per unit **increases** at high production levels.
- **Example:**
 - A **huge company** has **too many managers**, causing inefficiencies.

7. Case Study: Should You Expand Your Business?

Scenario:

- You own a **bakery**.
- Demand for cakes **increases**.
- Should you **hire more workers or buy new machines**?

Economic Analysis:

1. **Opportunity Cost** – Compare cost of hiring vs. investing in machines.
2. **Marginal Cost Analysis** – Will additional workers increase efficiency?
3. **Economies of Scale** – Will bulk purchasing reduce costs?

Decision:

- If **cost per unit decreases**, **expand**.
- If **marginal costs exceed revenue**, **do not expand**.

8. Accounting Cost (Explicit Cost) 📊

- **Definition:** Accounting cost includes **only actual expenses** incurred by a business. These are **direct, out-of-pocket costs** recorded in financial statements.

- **Formula:** $\text{Accounting Cost} = \text{Total Explicit Costs}$

$\text{Accounting Cost} = \text{Total Explicit Costs}$

- **Examples:**

- ✓ Salaries paid to employees 💰
- ✓ Rent for office space 🏢
- ✓ Cost of raw materials 🏭
- ✓ Utility bills (electricity, water) 💡
- ✓ Taxes paid to the government 📄

- **Used for:**

- ✓ **Financial reporting** and **tax calculations**.
- ✓ Shows **actual profit** but **ignores opportunity costs**.

9. Economic Cost (Explicit + Implicit Costs) 📈

- **Definition:** Economic cost includes both explicit costs and implicit (opportunity) costs.

- **Formula:** $\text{Economic Cost} = \text{Explicit Costs} + \text{Implicit Costs}$

$\text{Economic Cost} = \text{Explicit Costs} + \text{Implicit Costs}$

- **Implicit Costs = Opportunity costs** of using resources in one activity instead of another.

- **Examples:**

- ✓ **Lost salary:** If you quit a ₹10 lakh/year job to start a business, that lost salary is an **implicit cost**.
- ✓ **Using owned office space:** If you own a building but use it for your business instead of renting it out, the **foregone rent** is an **implicit cost**.
- ✓ **Investor's capital return:** If you invest ₹1 crore in a business instead of earning **5% interest in a bank**, the lost ₹5 lakh per year is an **implicit cost**.

- **Used for:**

- ✓ **Business decision-making** and **long-term planning**.
- ✓ Helps understand the **true profitability** of a business.

Final Takeaways

- ✓ **Opportunity Cost is key in business decision-making.**
- ✓ **Sunk Costs should be ignored in future decisions.**
- ✓ **Understanding cost structures helps in pricing and production planning.**
- ✓ **Economies of Scale reduce costs, but too much expansion can cause diseconomies.**

Week 2: Opportunity Cost and Cost Concepts

1. Opportunity Cost

Opportunity Cost = Value of Best Alternative Forgone

2. Marginal Cost

$$MC = \frac{\Delta TC}{\Delta Q}$$

- ΔTC = Change in Total Cost
- ΔQ = Change in Quantity

3. Total Cost

$$TC = FC + VC$$

- FC = Fixed Cost
- VC = Variable Cost

4. Average Total Cost (ATC)

$$ATC = \frac{TC}{Q}$$

5. Average Variable Cost (AVC)

$$AVC = \frac{VC}{Q}$$

Week 3

1. Marginal Rate of Technical Substitution (MRTS)

Definition

- The **rate at which one input (e.g., labor) can be substituted for another input (e.g., capital)** while keeping output constant.
- **Formula:**

$$MRTS = -(MP_{I1} / MP_{I2})$$

- **MP_{I1}** = Marginal product of Input 1.
- **MP_{I2}** = Marginal product of Input 2.

Example

- A factory uses **machines (K) and workers (L)**.
- If **one machine is removed**, but **two extra workers compensate** to maintain the same output:

$$MRTS = 2$$

- **Interpretation: 2 workers replace 1 machine.**

2. Isoquants: Representation of MRTS

What are Isoquants?

- **Curves that represent combinations of inputs** that produce the same output.
- **Similar to indifference curves in consumer theory.**

Properties of Isoquants

1. **Downward sloping** – To maintain output, increasing one input requires reducing another.
2. **Convex to the origin** – Substitution becomes harder as one input increases.
3. **Higher Isoquants = Higher output.**

3. Returns to Scale

Types of Returns to Scale

Type	Definition	Example
Increasing Returns to Scale (IRS)	Doubling inputs, more than doubles the output	A software company hiring more developers
Decreasing Returns to Scale (DRS)	Doubling inputs, less than doubles the output	Overcrowding in a factory
Constant Returns to Scale (CRS)	Doubling inputs doubles output	Factory assembly line

Production Function Representation

$$Q=f(K,L)$$

- If $f(2K, 2L) > 2Q$, IRS.
- If $f(2K, 2L) = 2Q$, CRS.
- If $f(2K, 2L) < 2Q$, DRS.

4. Cost Minimization and Isoquants

- **Goal:** Find the **cheapest combination** of inputs that achieves a desired output.
- **Key Condition for Cost Minimization:**

$$MP_L/w=MP_K/r$$

- **w** = Wage rate (cost of labor).
- **r** = Cost of capital (e.g., machine rental).

Example

- A firm has **two choices**:
 1. **More labor, fewer machines.**
 2. **More machines, fewer workers.**
- If labor is **cheaper than machines**, hire more **workers**.

5. Long-Run Cost Curves

Long-Run Total Cost (LRTC)

$$\text{LRTC} = wL + rK$$

- LRTC is derived from cost-minimizing input combinations.

Long-Run Average Cost (LRAC)

$$\text{LRAC} = \text{LRTC} / Q$$

- LRAC is U-shaped due to economies and diseconomies of scale.

Long-Run Marginal Cost (LRMC)

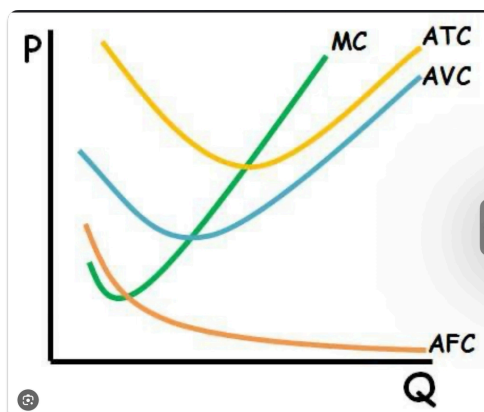
$$\text{LRMC} = \Delta \text{LRTC} / \Delta Q$$

6. Economies and Diseconomies of Scale in the Long Run

Concept	Definition	Example
Economies of Scale	Cost per unit decreases as production increases	Bulk purchasing of materials
Diseconomies of Scale	Cost per unit increases at high production levels	Too many employees reducing efficiency

7. Cost Curves and Their Relationships

Curve	Formula	Relation
MC (Marginal Cost)	$\text{MC} = \Delta \text{TC} / \Delta Q$	Intersects ATC and AVC at their minimums
ATC (Average Total Cost)	$\text{ATC} = \text{TC} / Q$	$\text{ATC} = \text{AVC} + \text{AFC}$
AVC (Average Variable Cost)	$\text{AVC} = \text{VC} / Q$	Always below ATC
AFC (Average Fixed Cost)	$\text{AFC} = \text{FC} / Q$	Always decreasing



8. Cost Minimization in Hiring Decisions

Problem:

- A firm is **hiring workers** and using **machines**.
- **Labor cost:** ₹10 per hour.
- **Machine cost:** ₹50 per hour.
- **MPL (Marginal Product of Labor)** = 10 units.
- **MPK (Marginal Product of Capital)** = 50 units.

Solution: Use Cost-Minimization Rule

$$MPL/w = MPK/r$$

$$10/10 = 50/50$$

- **Conclusion:** The firm is **optimally using labor and machines**.
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Final Takeaways

- ✓ MRTS explains how inputs can be substituted while keeping output constant.
- ✓ Isoquants show all input combinations that produce the same output.
- ✓ Cost minimization requires balancing labor and capital costs.
- ✓ Long-run cost curves determine economies/diseconomies of scale.

GAs FYI

Understanding the Production Technology in the Question

The key statement in the question is:

- ✓ "A firm can always trade three units of labour for one unit of capital and still keep output constant."

This means:

- Labour and capital can be **perfectly substituted** at a constant rate.
 - The production function must be **linear** in nature.
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Analyzing the Given Options

1. Cobb-Douglas Production Function ❌

- **Formula:**

$$Q = AL^\alpha K^\beta$$

- **Characteristics:**

- Output depends on both **labour (L)** and **capital (K)** in a **multiplicative** form.
- There is **diminishing marginal returns** to inputs.
- **Labour and capital are not perfectly substitutable.**

- **Why Incorrect?**

- In Cobb-Douglas, **capital and labour are complementary** rather than **perfect substitutes**.

1. Leontief Production Function ❌

- **Formula:**

$$Q = \min(aL, bK)$$

- **Characteristics:**

- **Fixed proportion** of capital and labour.
- No substitution is possible (if labour increases but capital is fixed, output remains unchanged).

- **Why Incorrect?**

- This function assumes that capital and labour must be used in a **fixed ratio** with **no substitution**.
- The question states that **labour can be traded for capital at a fixed rate**, implying **substitutability**.

1. Linear Production Function ✅ (Correct Answer)

- **Formula:**

$$Q = aL + bK$$

- **Characteristics:**

- Labour and capital can be **substituted at a constant rate**.
- If the firm can always trade **3 units of labour for 1 unit of capital**, the function could be written as:

$$Q = aL + bK \quad \text{where } b = 3a$$

- **Why Correct?**

- A linear production function allows **perfect substitutability** between inputs, which matches the problem statement.

Final Answer:

✅ **Linear Production Function** 🚀

Key Takeaways for Notes

✅ **Cobb-Douglas:** Inputs are complementary and have diminishing returns. ❌

✅ **Leontief:** Fixed proportion, no substitution. ❌

✅ **Linear:** Perfect substitution of inputs at a constant rate. ✅ (Correct Answer)

Let me know if you need further clarification! 🚀

The **Marginal Rate of Technical Substitution (MRTS)** is given by:

$$\text{MRTS} = \text{MPL} / \text{MPK}$$

Where:

- **MPL = 50 kg/hr** (Marginal Product of Labour)
- **MRTS=1/4** (Labour to Capital substitution rate)
- **MPK** = Marginal Product of Capital (to be found)

Step 1: Solve for MPK

$$\text{MPL} / \text{MPK} = 1/4$$

Final Answer:

✅ 200

I'll manually solve this step-by-step using the **Cost-Minimization Condition and Production Function**.

Step 1: Given Data

- **Production Function:**
 $q = 10L^{1/2}K^{1/2}$
- **Cost of Labour (w)** = ₹20 per unit
- **Cost of Capital (r)** = ₹80 per unit
- **Output (q)** = 100 units

Step 2: Cost-Minimization Condition

The **cost-minimizing combination of labour and capital** occurs when:

$$\text{MRTS} = \text{MPL} / \text{MPK} = w / r$$

Find Marginal Products

- **Marginal Product of Labour (MPL):**
 $\text{MPL} = \partial q / \partial L = 10 \times \frac{1}{2} L^{-1/2} K^{1/2} = 5 K^{1/2} L^{-1/2}$
- **Marginal Product of Capital (MPK):**
 $\text{MPK} = \partial q / \partial K = 10 \times \frac{1}{2} L^{1/2} K^{-1/2} = 5 L^{1/2} K^{-1/2}$

Setting **MRTS = w/r**:

$$\frac{\text{MPL}}{\text{MPK}} = \frac{5K^{1/2}L^{-1/2}}{5L^{1/2}K^{-1/2}} = \frac{K}{L} = \frac{20}{80} = \frac{1}{4}$$

$$K = \frac{1}{4}L$$

Step 3: Solve for LL and KK

Substituting into the **production function**:

$$100 = 10(L^{1/2})(K^{1/2}) \Rightarrow 100 = 10(L^{1/2})(K^{1/2})$$

$$10(L^{1/2})(L/4)^{1/2} = 100 \Rightarrow 10(L^{1/2})(L^{1/2}/2) = 100$$

$$10 \times L^{1/2} \times (L^{1/2}/2) = 100 \Rightarrow 10 \times L^{1/2} \times L^{1/2} / 2 = 100$$

$$10 \times (L/2) = 100 \Rightarrow 10 \times (L/2) = 100$$

$$5L = 100 \Rightarrow L = 20$$

$$L = 20 \Rightarrow L = 20$$

$$K = L/4 = 20/4 = 5 \Rightarrow K = \frac{L}{4} = \frac{20}{4} = 5$$

Step 4: Correct Answer

✓ **Optimal Labour (LL) = 20, Optimal Capital (KK) = 5**

✓ **Final Answer: (20, 5)**

Thus, the correct option is: **20, 5** 🚀

Key Takeaways for Notes

✓ **Cost-minimization occurs where MRTS = w/r**

✓ **MRTS is derived from the Marginal Products:**

$$MPL/MPK = \frac{w}{r} \Rightarrow \frac{MP_L}{MP_K} = \frac{w}{r}$$

✓ **For $q = 10L^{1/2}K^{1/2}$ with $w = 20$, $r = 80$, $q = 100$:**

- **Labour = 20 units, Capital = 5 units.**

✓ **Correct answer is (20, 5).**

Let me know if you need further clarifications! 🚀

I'll manually solve this step-by-step using the **production function**.

Step 1: Given Data

- **Production Function:**

$$q = 10L^{1/2}K^{1/2} \Rightarrow q = 10L^{1/2}K^{1/2}$$

- **Fixed Capital ($K=5$)**
- **New Output Target ($q=140$)**

Step 2: Solve for LL

$$140 = 10L^{1/2}(5)^{1/2} \Rightarrow 140 = 10L^{1/2}(5^{1/2})$$

Since $5^{1/2} \approx 2.236$:

$$140 = 10L^{1/2} \times 2.236 \Rightarrow 140 = 10L^{1/2} \times 2.236$$

$$140 = 22.36 L^{1/2} \Rightarrow L = \frac{140^2}{22.36} = 39.2$$

$$L^{1/2} = \frac{140}{22.36} = 6.26 \Rightarrow L = (6.26)^2 = 39.2$$

$$L = (6.26)^2 = 39.2$$

$$L = 39.2$$

Final Answer:

✓ **Labour required = 39.2 units** 🚀

Key Takeaways for Notes

✓ **With fixed capital, more labour is needed to increase output.**

✓ **For $q=140$, the firm needs $L=39.2$ units.**

Let me know if you need further clarifications! 🚀

I'll manually solve this step-by-step using the **Total Cost Formula**.

Step 1: Given Data

- **Cost of Labour (ww) = ₹20 per unit**
- **Cost of Capital (rr) = ₹80 per unit**
- **Labour Required (LL) = 39.2 units** (from previous question)
- **Fixed Capital (KK) = 5 units**

Step 2: Compute Total Cost

The **Total Cost (TC) formula**:

$$TC = (w \times L) + (r \times K)$$

$$TC = (20 \times 39.2) + (80 \times 5)$$

$$TC = 784 + 400$$

$$TC = 1,184$$

Final Answer:

✓ **Total cost of production = ₹1,184** 🚀

Key Takeaways for Notes

✓ **Total cost formula:**

$$TC = (w \times L) + (r \times K)$$

✓ **For $L=39.2$, $K=5$, $w=20$, $r=80$:**

- **Total cost = ₹1,184.**

Let me know if you need further clarifications! 🚀

Week 3: Production and Cost Minimization

1. Marginal Rate of Technical Substitution (MRTS)

$$MRTS = -\frac{\Delta K}{\Delta L} = \frac{MP_L}{MP_K}$$

- MP_L = Marginal Product of Labor
- MP_K = Marginal Product of Capital

2. Cost Minimization Rule

$$\frac{MP_L}{w} = \frac{MP_K}{r}$$

- w = Wage rate (cost of labor)
- r = Rental rate of capital

3. Long-Run Total Cost (LRTC)

$$LRTC = wL + rK$$

4. Long-Run Average Cost (LRAC)

$$LRAC = \frac{LRTC}{Q}$$

5. Long-Run Marginal Cost (LRMC)

$$LRMC = \frac{\Delta LRTC}{\Delta Q}$$

Week 4

1. Cost Curve Analysis

Short-Run vs. Long-Run Costs

- **Short Run:** Some inputs are **fixed** (e.g., factory, machines).
- **Long Run:** All inputs can be **adjusted** (e.g., new factory, better technology).

Fixed Cost vs. Variable Cost

Cost Type	Definition	Example
Fixed Cost (FC)	Costs that remain constant	Factory rent, salaries
Variable Cost (VC)	Costs that change with production	Raw materials, electricity
Total Cost (TC)	Sum of FC and VC	$TC = FC + VC$

Long-Run and Short-Run Cost Curves

- **Short-Run Cost Curve:** U-shaped due to **diminishing marginal returns**.
- **Long-Run Cost Curve:** Envelops multiple short-run curves, forming a **smooth U-shape**.

2. Consumer Behavior: Budget Constraints & Preferences

1. Building Blocks of Consumer Behavior

- **Consumption Possibility Set:** All **bundles** a consumer can afford.
 - **Consumption Feasibility Set:** Bundles **restricted by constraints** (legal, geographical, financial).
 - **Preferences:** Consumer choices based on rationality.
 - **Behavioral Assumption:** Consumers act **rationally** to maximize utility.
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3. Budget Sets and Budget Lines

Budget Constraint Formula

$$p_1 x_1 + p_2 x_2 = m$$

- **p_1, p_2** = Prices of goods.
- **x_1, x_2** = Quantities of goods.
- **m** = Consumer's income.

Effect of Price & Income Changes

Change	Impact on Budget Line
Increase in Income	Shifts outward (can afford more)
Decrease in Income	Shifts inward (can afford less)
Increase in Price of Good 1	Budget line tilts inward
Increase in Price of Good 2	Budget line tilts inward

4. Taxation, Rationing, and Coupons

- **Tax:** Reduces consumer purchasing power.
- **Rationing:** Limits maximum consumption of goods.
- **Coupons:** Allow specific quantities at reduced prices.

Example: If the government imposes a **quantity tax on Good 1** and **subsidizes Good 2**, the new budget equation becomes:

$$(p_1 + t) x_1 + (p_2 - s) x_2 = m - u$$

- **t** = Tax per unit.
 - **s** = Subsidy per unit.
 - **u** = Lump-sum tax.
-

5. Consumer Preferences and Utility

1. Rationality Assumptions

- **Completeness:** Consumers **can compare** any two goods.

- **Transitivity:** If A is preferred to B, and B is preferred to C, then A is preferred to C.

2. Utility Function

- Represents consumer preferences numerically.
- Higher utility → More preferred bundle.

3. Indifference Curves

- **Graph showing combinations of goods that give the same satisfaction.**
 - **Properties:**
 1. **Downward sloping** – If you consume more of one good, you must give up another.
 2. **Convex to the origin** – Diminishing Marginal Rate of Substitution (MRS).
-

6. Marginal Rate of Substitution (MRS)

Definition

- **Rate at which a consumer is willing to trade one good for another while maintaining the same utility.**
- **Formula:**

$$MRS = -MU_1 / MU_2$$
 - **MU1** = Marginal utility of good 1.
 - **MU2** = Marginal utility of good 2.

Example

- A consumer has **MRS = 2**, meaning they would give up **2 units of Good 2** for **1 unit of Good 1**.
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7. Monotonic Preferences and Convexity

1. Monotonicity of Preferences

- **Definition:** More is always **better**.
- **Mathematical Representation:**
 - If $(x_1, x_2) \leq (y_1, y_2)$ and $(y_1, y_2) \neq (x_1, x_2)$
 - Then, (y_1, y_2) is **preferred** to (x_1, x_2) .

2. Convexity of Preferences

- **Consumers prefer balanced consumption.**
 - **Example:**
 - A consumer prefers **1 apple & 1 banana** over **2 apples & 0 bananas**.
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8. Graded Assignment Solutions (Key Takeaways)

1. Cost Minimization in Production

Scenario: A company has **three plants** with different cost functions:

$$TC_1 = 300 - 10q + 50q^2 \quad TC_1 = 300 - 10q + 50q^2$$

$$TC_2 = 50 + 10q^2 \quad TC_2 = 50 + 10q^2$$

$$TC_3 = 1000 + 20q \quad TC_3 = 1000 + 20q$$

- **Solution:**
 - If producing **5 units**, the cheapest option is **Plant 2 (Cost = ₹300)**.

2. Budget Line with Taxes & Subsidies

- **New budget equation after tax/subsidy:**
$$(p_1 + t)x_1 + (p_2 - s)x_2 = m - u \quad (p_1 + t)x_1 + (p_2 - s)x_2 = m - u$$
- **Interpretation:**
 - **Taxes** increase effective price.
 - **Subsidies** decrease effective price.
 - **Lump-sum tax (u)** reduces overall income.

3. Consumer Well-being and Income Changes

- **If income increases and one price falls**, the consumer is **at least as well-off** because:
 - They can afford **all previous bundles** and possibly more.

4. Indifference Curve Properties

- **Psychological Assumption 1: Non-Satiation** – More is better.
- **Psychological Assumption 2: Convexity** – Balanced consumption is preferred.

5. Managerial Decision on Expanding a Factory

- **Optimal Decision:**
 - Compare **incremental benefit from expansion to incremental cost**.
 - **Correct approach:**
$$\text{Marginal Benefit} > \text{Marginal Cost} \quad \text{Marginal Benefit} > \text{Marginal Cost}$$

6. Marginal Analysis in Hiring

- **MR (Marginal Revenue) = $100 \times 12 = ₹1200$**
- **MC (Marginal Cost) = ₹1500**
- Since **MR < MC**, the firm should **not hire the worker**.

7. Budget Line and Slope Interpretation

- **Equation:**
$$20C + 10F = 1000 \quad 20C + 10F = 1000$$
- **Slope:** (Negative ratio of prices).
$$-2/2$$

- **Intercepts:**
 - X-axis: (Coconut).
5050
 - Y-axis: (Fish).
100100

Final Takeaways

- ✓ Cost curves help firms decide short-run vs. long-run production.
- ✓ Budget lines show consumer constraints, and their slope is determined by relative prices.
- ✓ Taxes and subsidies alter budget constraints.
- ✓ MRS explains how consumers substitute goods to maintain utility.
- ✓ Monotonicity & convexity ensure rational decision-making.

Week 4: Consumer Behavior and Utility

1. Budget Constraint Equation

$$p_1x_1 + p_2x_2 = m$$

- p_1, p_2 = Prices of goods
- x_1, x_2 = Quantities of goods
- m = Consumer's income

2. Tax and Subsidy Adjusted Budget Constraint

$$(p_1 + t)x_1 + (p_2 - s)x_2 = m - u$$

- t = Tax per unit
- s = Subsidy per unit
- u = Lump-sum tax

3. Marginal Rate of Substitution (MRS)

$$MRS = -\frac{MU_1}{MU_2}$$

- MU_1 = Marginal Utility of Good 1
- MU_2 = Marginal Utility of Good 2

4. Indifference Curve Slope

$$MRS = \frac{p_1}{p_2}$$

- The consumer optimally consumes where **MRS = price ratio**.

Week 5

1. The Consumer's Problem

A consumer aims to **maximize utility** subject to a budget constraint.

Four Building Blocks of Consumer Choice

1. **Consumption Set** – All possible bundles a consumer can choose.
2. **Feasible Set** – Bundles the consumer can afford within their budget.
3. **Preferences → Utility Function** – A function that assigns a satisfaction level to each bundle.
4. **Consumer's Problem** – Choosing the best bundle given income and prices.

Utility Maximization

$\max U(x_1, x_2) \text{ subject to } p_1 x_1 + p_2 x_2 \leq M$
 $\max U(x_1, x_2) \quad \text{subject to} \quad p_1 x_1 + p_2 x_2 \leq M$

where:

- x_1, x_2 = quantities of goods.
- p_1, p_2 = prices of goods.
- M = consumer's income.

Marginal Utility of Income

- Measures how additional income impacts utility.
- **Diminishing marginal utility**: As income increases, additional utility gained per unit of money decreases.

Corner Solution

Occurs when a consumer **spends all income on one good**, ignoring the other.

- Example: If good 2 is too expensive relative to good 1, **only good 1 is purchased**.

2. Preferences and Utility Functions

Consumers have different preferences, leading to different types of utility functions.

Types of Preferences

1. **Perfect Substitutes**
 - Consumer is willing to exchange one good for another at a constant rate.
 - **Utility Function**:
 $U(x_1, x_2) = ax_1 + bx_2$
 - **Example**: Pepsi vs. Coke.

2. Perfect Complements

- Goods are consumed together in **fixed proportions**.

- **Utility Function:**

$$U(x_1, x_2) = \min(ax_1, bx_2) \quad U(x_1, x_2) = \min(ax_1, bx_2)$$

- **Example:** Left and right shoes.

3. Cobb-Douglas Preferences

- Well-balanced trade-off between goods.

- **Utility Function:**

$$U(x_1, x_2) = x_1^a x_2^b \quad U(x_1, x_2) = x_1^a x_2^b$$

- **Example:** Buying food and rent.

4. Bliss Point Preferences

- Consumer has a **most preferred bundle**, beyond which more consumption **lowers utility**.

5. Quasi-Linear Preferences

- One good provides linear utility, while the other provides diminishing returns.

- **Utility Function:**

$$U(x_1, x_2) = f(x_1) + x_2 \quad U(x_1, x_2) = f(x_1) + x_2$$

3. Demand Functions

A **demand function** shows how quantity demanded changes with price, income, and other factors.

Marshallian Demand Function

$$x_i = f(p_1, p_2, M) \quad x_i = f(p_1, p_2, M)$$

- **Graphically:** Demand curves slope **downward** (law of demand).

Effects of Price Changes

- **Substitution Effect:** Consumers switch to a cheaper alternative.
- **Income Effect:** A price drop increases real income, allowing consumers to buy more.

Effect of Income Changes

- **Normal Goods:** Demand increases as income rises.
- **Inferior Goods:** Demand decreases as income rises.

4. Demand, Supply, and Market Equilibrium

Consumer Surplus

- **Consumer's Surplus:** The difference between what a consumer **is willing to pay** and what they actually pay.
- **Graphically:** The area **above the price line** and **below the demand curve**.

Supply Function

- **Factors Affecting Supply:**

- Input costs.
- Technology.
- Government policies.
- Number of sellers.

- **Market Supply Function:**

$$Q_S = f(p, w, T, N)$$

where:

- p = price of the good.
- w = wages.
- T = technology.
- N = number of firms.

Market Equilibrium

$$Q_D = Q_S$$

- **At equilibrium:**

- No shortage or surplus.
- The market clears.

Deadweight Loss (DWL)

- Occurs when markets **don't operate efficiently** due to price controls, taxes, or monopolies.
- **Graphically:** The area of lost surplus in a **distorted market**.

Changes in Market Equilibrium

- **Increase in Demand** → Higher price & quantity.
- **Increase in Supply** → Lower price & higher quantity.

Key Takeaways

- Consumers maximize utility given income constraints.
 - Different utility functions represent different consumer preferences.
 - Demand functions show how price and income changes affect quantity demanded.
 - Market equilibrium is achieved when supply equals demand.
 - Consumer & producer surpluses measure market efficiency.
 - Deadweight loss occurs when markets are inefficient.
-

Week 5 - Consumer Theory and Demand Functions

1. Budget Constraint Equation

$$p_1 x_1 + p_2 x_2 = M$$

- x_1, x_2 = quantities of goods.
- p_1, p_2 = prices of goods.
- M = consumer's income.

2. Marginal Rate of Substitution (MRS)

$$MRS = -\frac{MU_{x_1}}{MU_{x_2}}$$

- Shows the rate at which consumers are willing to **substitute one good for another**.

3. Marshallian Demand Function

$$x_i = f(p_1, p_2, M)$$

4. Consumer Surplus (CS)

$$CS = \int_0^Q P dQ - P \times Q$$

- Difference between **willingness to pay** and **actual price paid**.

Week 6

Topics: Elasticity, Market Interventions, and Taxation

1. Elasticity in Economics

Elasticity measures how **responsive** quantity demanded or supplied is to changes in price, income, or other factors.

Types of Elasticity

1.1. Price Elasticity of Demand (PED)

$$PED = \frac{\Delta Q_d}{Q_d} \div \frac{\Delta P}{P} \text{ or } PED = \frac{\Delta Q_d}{Q_d} \times \frac{P}{\Delta P}$$

- If $PED > 1$ → **Elastic** (luxury goods).
- If $PED < 1$ → **Inelastic** (necessities, medicine).
- If $PED = 1$ → **Unit elastic** (proportional response).

Example PED values:

- **Cigarettes (US):** 0.3 to 0.6 (inelastic).
- **Wine:** 1.0 (unit elastic).
- **Airline tickets (pleasure travel):** 1.5 (elastic).

1.2. Point vs. Arc Elasticity

- **Point Elasticity:** Used for small changes in price.

$$PED = \frac{dQ}{Q} \div \frac{dP}{P} \text{ or } PED = \frac{dQ}{dP} \times \frac{P}{Q}$$

- **Arc Elasticity:** Used when there is a large price change.

$$PED = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \div \frac{P_2 - P_1}{(P_1 + P_2)/2} \text{ or } PED = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \times \frac{(P_1 + P_2)/2}{P_2 - P_1}$$

1.3. Factors Affecting Price Elasticity of Demand

1. **Availability of Substitutes** – More substitutes → higher elasticity.
2. **Budget Share** – If a good is a large part of income → higher elasticity.
3. **Time Horizon** – Demand is more elastic in the long run.
4. **Definition of the Good** – Broader categories (e.g., food) have lower elasticity than specific brands.

1.4. Cross-Price Elasticity of Demand

Cross-Elasticity = $\frac{\Delta Q_A}{\Delta P_B}$ $\text{Cross-Elasticity} = \frac{\Delta Q_A}{\Delta P_B}$

- **If positive** → **Substitutes** (tea vs. coffee).
- **If negative** → **Complements** (cars and petrol).

1.5. Income Elasticity of Demand

Income Elasticity = $\frac{\Delta Q_d}{\Delta M}$ $\text{Income Elasticity} = \frac{\Delta Q_d}{\Delta M}$

- **If $E > 1$** → **Luxury good** (automobiles: 2.98).
- **If $0 < E < 1$** → **Normal good** (energy: 0.7).
- **If $E < 0$** → **Inferior good** (public transport: -0.36).

1.6. Advertisement Elasticity of Demand

Measures how demand changes with **advertising expenditures**.

AE = $\frac{\Delta Q_d}{\Delta A}$ $\text{AE} = \frac{\Delta Q_d}{\Delta A}$

1.7. Price Elasticity of Supply (PES)

PES = $\frac{\Delta Q_s}{\Delta P}$ $\text{PES} = \frac{\Delta Q_s}{\Delta P}$

- **Key Factors Affecting PES:**
 - **Availability of factors of production.**
 - **Production process flexibility.**
 - **Inventories and storage capability.**
 - **Time horizon (short-run vs. long-run).**

2. Market Interventions

2.1. Taxes and Their Impact

A **tax** shifts the supply curve **upward** by the amount of the tax.

2.2. Tax Incidence (Who Bears the Burden?)

Tax Incidence = $\frac{PED}{PED + PES}$ $\text{Tax Incidence} = \frac{\text{PED}}{\text{PED} + \text{PES}}$

- **If demand is inelastic** → Buyers bear most of the tax burden.
- **If supply is inelastic** → Sellers bear most of the tax burden.

2.3. Ad Valorem vs. Specific Taxes

- **Ad Valorem Tax:** Percentage of the price (e.g., VAT).
- **Specific Tax:** Fixed amount per unit sold (e.g., ₹5 per liter of petrol).

2.4. Welfare Effects of a Tax

- **Consumer Surplus Decreases.**
 - **Producer Surplus Decreases.**
 - **Government Revenue Increases.**
 - **Deadweight Loss (DWL):** Efficiency loss due to tax distortions.
-

3. Price Controls and Market Interventions

3.1. Price Floors (Minimum Price)

- **Definition:** Government sets a **minimum price** above equilibrium.
- **Examples:** Minimum wage, agricultural price support.
- **Effects:**
 - **Surplus** (excess supply).
 - **Black markets** (firms sell below the official price).

3.2. Price Supports

- **Definition:** Government buys surplus goods to maintain prices.
- **Example:** Government buys excess wheat at a guaranteed price.

3.3. Production Quotas

- **Definition:** Government limits the amount that can be produced.
- **Example:** OPEC oil production limits.

3.4. Voluntary Production Reductions

- **Definition:** Firms reduce output voluntarily (often subsidized by the government).

3.5. Price Ceilings (Maximum Price)

- **Definition:** Government sets a **maximum price** below equilibrium.
 - **Examples:** Rent controls, essential medicine price caps.
 - **Effects:**
 - **Shortages** (excess demand).
 - **Black markets** (people willing to pay more).
-

4. Import Tariffs and Quotas

4.1. Tariffs

- **Definition:** A **tax** on imported goods.

- **Effects:**
 - Higher domestic prices.
 - Government earns revenue.
 - Foreign producers lose market share.

4.2. Quotas

- **Definition:** A restriction on the quantity of imports.
- **Effects:**
 - Same as tariffs, but without government revenue.
 - Increases domestic prices.

Key Takeaways

- Elasticity measures how demand/supply responds to changes in price, income, or advertising.
- Tax incidence depends on relative elasticities of demand and supply.
- Price floors cause surpluses, while price ceilings cause shortages.
- Tariffs and quotas restrict imports but raise domestic prices.

Week 6 - Elasticity, Taxation, and Market Interventions

5. Price Elasticity of Demand (PED)

$$PED = \frac{\% \Delta Q_d}{\% \Delta P}$$

6. Arc Elasticity Formula

$$PED = \frac{(Q_2 - Q_1)}{(Q_1 + Q_2)/2} \div \frac{(P_2 - P_1)}{(P_1 + P_2)/2}$$

7. Cross-Price Elasticity of Demand

$$E_{AB} = \frac{\% \Delta Q_A}{\% \Delta P_B}$$

- Positive: Substitutes
- Negative: Complements

8. Income Elasticity of Demand

$$E_I = \frac{\% \Delta Q_d}{\% \Delta M}$$

- $E_I > 1 \rightarrow$ Luxury good
- $0 < E_I < 1 \rightarrow$ Normal good
- $E_I < 0 \rightarrow$ Inferior good

9. Tax Incidence Formula

$$\frac{PED}{PED + PES}$$

- Determines whether consumers or producers bear the tax burden.

10. Deadweight Loss (DWL) Due to Tax

$$DWL = \frac{1}{2} \times \underset{\downarrow}{\text{Tax}} \times (Q_{\text{before}} - Q_{\text{after}})$$

Week 7

Topics: Demand Estimation and Monopoly

1. Demand Estimation Methods

Firms need to estimate demand functions to set **optimal prices and production levels**. There are several methods to do this:

1.1. Managerial Judgment

- Uses **expert opinions** (sales managers, product engineers).
- Based on past sales data and competitive pricing.
- **Example:** A firm uses **historical knowledge** to decide if raising prices will reduce demand.

1.2. Analogous Products

- Works well for **incremental innovations**.
- Compares new product demand with **previous versions**.
- **Example:** Predicting iPhone 15 sales based on iPhone 14 demand.

1.3. Benchmarking

- Compares product performance with **competitive products**.
- Works best when there are **existing competitors**.

1.4. Focus Groups

- Gathers consumer reactions to price and product changes.
- Useful for qualitative research.
- **Limitation:** **Sample bias** and **small group size** may misrepresent market reality.

1.5. Purchase Intention Surveys

- Consumers rate their **likelihood of purchasing a product** at different prices.
- Uses a **five-point scale** (e.g., Definitely Buy → Definitely Not Buy).

1.6. Experimentation

- **Field Experiments:** Test real-world reactions to price changes.
- **Simulations:** Computer models predicting demand changes.

1.7. Regression Analysis of Historical Data

- Uses **past data** to estimate how demand changes with price, advertising, and other factors.
- **Example:** A car manufacturer analyzes **past sales vs. gas prices**.

1.8. Conjoint Analysis

- Breaks a product into **attributes** and determines which features consumers value most.
- Helps businesses determine the **best pricing strategy**.

1.9. Economic Value to Customer (EVC)

- Calculates the **maximum price a customer would pay** by comparing the product to alternatives.
 - **Example:** A new electric car is priced based on **fuel savings over 5 years**.
-

2. Monopoly: Definition and Characteristics

A **monopoly** is a market where **one firm** dominates and sets prices without competition.

2.1. Monopoly vs. Perfect Competition

Feature	Perfect Competition	Monopoly
Number of Firms	Many	One
Pricing Power	None (Price-Taker)	High (Price-Maker)
Barriers to Entry	Low	High
Demand Curve	Perfectly elastic	Downward sloping

2.2. Monopoly Pricing Rule

A **monopolist maximizes profit** where:

$$MR = MC$$

where:

- MR = Marginal Revenue
- MC = Marginal Cost

If $MR > MC$, increase production.

If $MR < MC$, reduce production.

2.3. Demand Curve and Marginal Revenue

- A **monopolist faces a downward-sloping demand curve**.
- **To sell more units, the firm must lower the price.**

$$MR = P + \frac{dP}{dQ} Q$$

3. Welfare Implications of Monopoly

3.1. Deadweight Loss (DWL)

- Monopoly **restricts output** below the socially optimal level.
- Creates **DWL** because some consumers **willing to pay above cost** cannot buy.

3.2. Monopolist's Supply Decision

- Unlike competitive firms, **monopolies do NOT have a supply curve**.
- **Supply depends on the shape of the demand curve.**

3.3. Price Discrimination

A **monopoly can charge different prices** to maximize profits.

Types of Price Discrimination:

1. **First-Degree (Perfect Price Discrimination):**
 - Each consumer pays **exactly what they are willing to pay**.
 - **Example:** Auctions, car dealerships.
 2. **Second-Degree (Quantity-Based Pricing):**
 - Different prices for **different quantities**.
 - **Example:** Bulk discounts, airline tickets.
 3. **Third-Degree (Group-Based Pricing):**
 - Different prices for **different consumer groups**.
 - **Example:** Student discounts, senior citizen fares.
-

4. Sources of Monopoly Power

4.1. Economic Barriers

- **Exclusive control over resources** (e.g., De Beers diamonds).
- **Technological superiority** (e.g., Intel in microprocessors).
- **Economies of scale** → Leads to **natural monopolies**.
- **Capital requirements** (e.g., High startup costs in telecom).

4.2. Legal Barriers

- **Patents** grant **exclusive production rights** (e.g., Pharmaceutical companies).
 - **Government Licenses** restrict market entry (e.g., Electricity, Railways).
-

5. Taxation and Monopoly

- A **monopoly reacts to taxes differently** from competitive firms.
 - **Effects of a tax on a monopolist:**
 - **Increases cost, leading to higher prices.**
 - **Reduces consumer surplus.**
 - **Can increase deadweight loss.**
-

6. Real-World Monopolies and Competition Policy

6.1. Natural Monopolies

- **Utilities (Electricity, Water)** have **high fixed costs**.
- **Governments regulate prices** to prevent exploitation.

6.2. Anti-Monopoly Regulations

- **Antitrust laws** prevent firms from **abusing monopoly power**.
 - **Example:** The **U.S. vs. Microsoft (1998)** – Microsoft was accused of **restricting competition**.
-

Key Takeaways

- **Demand estimation is critical** for pricing strategy and involves various methods (managerial judgment, surveys, regression, etc.).
- **A monopolist sets prices strategically** where **MR = MC**.
- **Monopolies cause deadweight loss** by restricting output.
- **Price discrimination helps monopolists maximize profits** by charging different consumers differently.
- **Legal and economic barriers protect monopolies** but can also lead to government regulation.

Week 7 - Monopoly and Demand Estimation

11. Profit Maximization Condition for Monopoly

$$MR = MC$$

12. Marginal Revenue (MR) Formula

$$MR = P + \frac{dP}{dQ}Q$$

13. Price Markup Over Marginal Cost

$$P = \frac{MC}{1 - \frac{1}{|PED|}}$$

- Higher markup when demand is inelastic.

14. Herfindahl-Hirschman Index (HHI)

$$HHI = \sum_{i=1}^n s_i^2$$

- Measures market concentration.

15. Concentration Ratio (CR4)

$$CR4 = \sum \text{Market Share of Top 4 Firms}$$

Week 8

Topics: Price Discrimination, Monopsony, Imperfect Competition, and Game Theory

1. Price Discrimination

Price discrimination occurs when a firm **charges different prices to different consumers for the same product** without cost differences.

1.1. First-Degree Price Discrimination (Perfect Price Discrimination)

- **Definition:** Firm charges each consumer **the maximum price they are willing to pay**.
- **Example:** Auctions, personalized pricing in online shopping.
- **Effects:**
 - **No consumer surplus** (firm captures all gains).
 - **Higher profits**.
 - **Difficult to implement** due to consumer preference secrecy.

1.2. Second-Degree Price Discrimination (Quantity Discounts)

- **Definition:** Price depends on the **quantity purchased**.
- **Example:**
 - Bulk pricing: ₹100 per unit for the first 10 units, ₹80 per unit for the next 10.
 - Electricity tariffs: First 100 kWh at ₹5, next 100 kWh at ₹4.
- **Effects:**
 - **Encourages bulk buying.**
 - **Reduces per-unit costs for consumers.**

1.3. Third-Degree Price Discrimination (Market Segmentation)

- **Definition:** Prices vary **across consumer groups** based on elasticity.
 - **Example:**
 - **Student & Senior Discounts.**
 - **Higher airfares for business travelers.**
 - **Effects:**
 - **More profit from inelastic consumers.**
 - **Requires ability to segment markets.**
-

2. Monopsony (Single Buyer Market)

A **monopsony** is a market structure where **one buyer dominates**. This buyer has **market power** and can set wages/prices.

2.1. Key Features of Monopsony

- **Single dominant buyer** (e.g., government purchasing defense equipment).
- **Workers/sellers have few alternative buyers.**
- **Monopsony power leads to lower wages/output.**

2.2. Monopsony vs. Monopoly

Feature	Monopoly (Single Seller)	Monopsony (Single Buyer)
Market Power	Controls price by limiting supply	Controls price by restricting demand
Effect on Price	Higher than competitive level	Lower than competitive level
Example	Microsoft (Windows)	Amazon (controlling supplier prices)

2.3. Example: Monopsony in Labor Markets

- **Amazon & Walmart** as dominant employers in local markets.
 - **College sports (NCAA)** paying student-athletes **below market wages**.
 - **Effect: Lower wages and employment.**
-

3. Imperfect Competition: Oligopoly and Monopolistic Competition

Most real-world markets are neither **perfectly competitive** nor **pure monopolies**. Instead, they fall under **imperfect competition**.

3.1. Market Structures Based on Firm Number and Product Type

Market Type	Number of Firms	Product Differentiation
Perfect Competition	Many	Homogeneous
Monopolistic Competition	Many	Differentiated
Oligopoly	Few	Homogeneous or Differentiated
Monopoly	One	Unique

3.2. Key Features of Imperfect Competition

- **Oligopoly:** Few firms with interdependence (e.g., automobile industry).
- **Monopolistic Competition:** Many firms with **differentiated products** (e.g., restaurants, clothing brands).
- **Firms have price-setting power.**
- **Advertising and branding matter.**

3.3. Measuring Market Concentration

- **Four-Firm Concentration Ratio (4CR):** Share of total sales by the top 4 firms.
- **Herfindahl-Hirschman Index (HHI):**

where s_i is the market share of each firm.

$$HHI = \sum_{i=1}^n s_i^2$$

s_i

4. Game Theory and Strategic Behavior

Game theory analyzes **strategic decision-making** when one firm's action affects others.

4.1. Key Components of a Game

- **Players:** Participants in the market.
- **Strategies:** Actions available to players.
- **Payoffs:** Outcomes from different choices.
- **Equilibrium:** Stable outcome where no player wants to change their strategy.

4.2. Prisoner's Dilemma (Oligopoly Example)

Two firms (A & B) must decide whether to **collude (high prices)** or **compete (low prices)**.

Firm A/Firm B	High Price	Low Price
High Price	(5,5)	(0,7)
Low Price	(7,0)	(2,2)

- **Dominant Strategy:** Both firms choose **low prices**, even though both would benefit from high prices.
- **Real-World Example: Price wars between airlines.**

4.3. Nash Equilibrium

A **Nash equilibrium** occurs when **no player can improve their payoff by changing their strategy** alone.

No firm has an incentive to deviate. \text{No firm has an incentive to deviate.}

4.4. Battle of the Sexes (Coordination Game)

- Players prefer **different but mutually beneficial outcomes**.
- **Example: Choosing between two locations for a date.**
- **Real-world application:** Firms **choosing industry standards** (e.g., Blu-ray vs. HD-DVD).

4.5. Strategies in Competitive Markets

- **Best Response:** Choose the best strategy **given what the other player does**.
- **Mixed Strategies:** Randomized decisions (e.g., penalty kicks in soccer).
- **Repeated Games:** Cooperation can emerge over time (e.g., **tit-for-tat strategy** in business pricing).

Key Takeaways

- **Price discrimination** allows firms to extract maximum consumer surplus.
 - **Monopsony power** results in lower wages and fewer jobs.
 - **Market structures** vary from perfect competition to monopoly.
 - **Oligopolies** engage in strategic decision-making.
 - **Game theory** helps explain competition, collusion, and pricing wars.
 - **Nash equilibrium** predicts stable strategies where no firm wants to change its actions.
-

Week 8 - Price Discrimination, Monopsony, and Game Theory

16. First-Degree Price Discrimination Revenue Maximization

$$\text{Profit} = \sum_{i=1}^n (P_i - MC) \times Q_i$$

17. Monopsony Equilibrium Condition

$$MRP_L = MFC_L$$

- **MRP_L**: Marginal Revenue Product of Labor
- **MFC_L**: Marginal Factor Cost of Labor

18. Nash Equilibrium Condition

No player can improve their outcome by unilaterally changing their strategy.

19. Prisoner's Dilemma Payoff Matrix Condition

- **Dominant Strategy**: Always choose the best response **regardless of what the other player does**.

20. Best Response Function (for Strategic Firms)

$$BR_A(P_B) = \max \text{profit given } P_B$$

21. Tit-for-Tat Strategy in Repeated Games

- Start with cooperation, then repeat what the opponent did in the previous round.

Week 9

1. Game Theory II

Rules of the Game

- **Players**: Decision-makers in the game.
- **Strategies**: Feasible actions available to each player.
- **Payoffs**: Outcomes or rewards from strategy combinations.

Game Structure

- **Timing of Moves**: Sequential or simultaneous.
- **Nature of Conflict**: Strategic interaction.
- **Information**:
 - **Perfect**: Players know full history.
 - **Complete**: Everyone knows payoffs/strategies.
- **Enforceability**: Whether agreements can be enforced.

Assumptions

- Rationality
- Common knowledge

Important Concepts

- **Dominant Strategy Equilibrium**: A strategy that is best regardless of what the opponent does.

- **Nash Equilibrium:** No player can improve their payoff by changing strategy unilaterally.
- **Subgame Perfect Equilibrium:** Uses **backward induction** to remove non-credible threats.

Extensive Form Game

- Represented as a **tree structure**.
- Nodes = Decision points
- Branches = Actions
- One starting point, no cycles.

Example – Entry Game (Coke vs Pepsi)

	Fight	Accommodate
Enter	-2, -1	1, 2
Out	0, 5	0, 5

- Using **backward induction**, the optimal choice is **Accommodate**.
- Shows **importance of credibility of threats**.

2. Classification of Games

Type	Description
Non-Cooperative	Players act independently, strategies detailed.
Cooperative	Outcomes described without individual strategies.
Static vs Dynamic	One-shot vs sequential decisions.
Perfect Info	History known to all players.
Complete Info	Payoffs/strategies known to all.
Incomplete Info	Some elements unknown to players.

3. Oligopoly

Definition

- Few firms dominate the market.
- Strategic interdependence among firms.

Models of Oligopoly

1. **Cournot Competition:**
 - Firms choose **quantity** simultaneously.
 - Each firm's output affects market price.
2. **Bertrand Competition:**
 - Firms compete by setting **prices**.
 - Price war drives prices to marginal cost under perfect substitutes.

3. **Stackelberg Competition:**
 - Sequential quantity setting.
 - One firm (leader) moves first, other (follower) responds.
 4. **Collusion & Cartels:**
 - Firms may collude to restrict output and raise price.
 - Illegal in many countries.
 5. **Locational Models:**
 - Firms compete on location in addition to price.
-

4. Monopolistic Competition

Features

- Many firms (but fewer than perfect competition).
- **Product differentiation:** Key feature (branding, quality, etc.).
- **Free entry and exit** in the long run.

Firm's Demand Curve

- **Downward sloping** because of differentiated products.
- Some pricing power (unlike perfect competition).

Long Run Dynamics

- Entry of new firms reduces demand faced by each existing firm.
- **Zero economic profit** in the long run, but still:
 - **Excess capacity**
 - **Markup pricing**

Welfare Implications

- **Less efficient** than perfect competition.
 - Consumers benefit from variety, but at a cost:
 - Higher prices
 - Duplication of fixed costs
-

◆ WEEK 9 – Game Theory, Oligopoly, Monopolistic Competition

◆ Payoff Matrix

- Player A & B:

$$\text{Payoff}_{A,B} = (\text{A's outcome}, \text{B's outcome})$$

◆ Best Response

- Choose strategy that gives **highest payoff** given the opponent's strategy.

◆ Nash Equilibrium

- No player can benefit by changing strategy **unilaterally**.

◆ Cournot Model (Oligopoly)

- Firms choose **quantity**, price is set by market demand.
- Reaction function:

$$q_1 = \frac{a - c - q_2}{2}$$

◆ Bertrand Model

- Firms choose **price**; equilibrium price = marginal cost if identical goods.

Week 10

1. Advertisement as a Signaling Device

- Firms use **advertising** not just to inform but to **signal quality**.
 - **High-quality firms** advertise more because:
 - They recover costs via repeat sales.
 - For **low-quality firms**, ad spending is not worthwhile.
 - Hence, advertising can be a **credible signal** of quality in **asymmetric information** markets.
-

2. Risk vs. Uncertainty

- **Risk**: Outcomes are **known** with associated **probabilities**.
 - **Uncertainty**: Outcomes or probabilities are **not known**.
-

3. Measuring Risk:

- **Probability**: Objective or subjective chance of an event.
- **Expected Value (EV)**: Average outcome weighted by probabilities.
- **Variance (σ^2)**: Measures spread of outcomes.
- **Standard Deviation (σ)**: Square root of variance.
- **Coefficient of Variation (CV)**:

$$CV = \sigma / E(X)$$

(Relative risk measure)

4. Decision Rules Under Risk

Rule	Description
1. Maximize EV	Choose option with highest expected value.
2. Mean-Variance Analysis	Prefer higher EV and lower variance.
3. Coefficient of Variation	Choose option with lowest CV when EVs are similar.

5. State-Contingent Decisions

- Contracts can depend on **state of nature**.
- **State-contingent contract**: Only triggers if a specific outcome occurs (e.g., insurance pays only during crop failure).

6. Expected Utility Theory (EUT)

- People maximize **expected utility**, not just expected value.
- EUT accommodates **risk preferences** via utility functions.

7. Risk Preferences

Type	Utility Curve	Behavior
Risk Averse	Concave	Prefers certain outcomes to risky ones.
Risk Neutral	Linear	Indifferent between certain and risky outcomes.
Risk Loving	Convex	Prefers risky outcomes to certain ones.

8. Insurance Concepts

- **Insurance Terms**:
 - **Premium**: Payment made to insurer.
 - **Benefit**: Compensation if insured event occurs.
 - **Actuarially Fair**: Premium = Expected payout.
- **Risk-averse individuals** are **more likely** to buy insurance.

9. Demand for Insurance

- Willing to pay premium if it reduces risk, even if premium > expected loss (less-than-fair insurance).
- Insurance is **valuable** as it **smooths consumption** across states.

10. Market for Insurance

- **Risk Pooling**: Combining uncorrelated risks (e.g., health insurance).
- **Risk Spreading**: Sharing risk across individuals or firms.
- **Risk Transfer**: Shift burden to insurer.

11. Decisions Under Uncertainty (No Probabilities)

Decision Criteria:

Rule	Logic
Maximax	Choose option with best possible outcome . (Optimistic)
Maximin	Choose option with best worst-case scenario . (Pessimistic)
Minimax Regret	Choose option that minimizes potential regret .
Equal Probability	Assume all states are equally likely.

Example Payoff Table:

Option	Normal	Dry	Flood
Usual Practice	5 Cr	-1 Cr	-3 Cr
More Fertilizer	3 Cr	2 Cr	0.5 Cr
More Insecticide	2 Cr	1 Cr	0.75 Cr

Different rules would suggest different “best” decisions.

◆ WEEK 10 – Risk, Insurance, Decision Under Uncertainty

◆ Expected Value (EV)

$$E(X) = \sum P_i \cdot X_i$$

◆ Variance & Standard Deviation

$$\text{Var}(X) = \sum P_i \cdot (X_i - E(X))^2 \quad ; \quad \sigma = \sqrt{\text{Var}(X)}$$

◆ Coefficient of Variation (CV)

$$\text{CV} = \frac{\sigma}{E(X)}$$

◆ Expected Utility

$$E[U(X)] = \sum P_i \cdot U(X_i)$$

◆ Certainty Equivalent (CE)

- Amount of certain income that yields same utility as expected utility.

◆ Risk Premium

$$\text{Risk Premium} = E(X) - \text{CE}$$

◆ Insurance Premium (Actuarially Fair)

$$\text{Premium} = \sum P_i \cdot \text{Loss}_i$$

◆ Decision Criteria Under Uncertainty

Rule	Formula
Maximax	Max(max payoff)
Maximin	Max(min payoff)
Minimax Regret	Min(max regret)
Equally Likely	Average of all outcomes

Week 11

1. Asymmetric Information: Overview

- **Definition:** When one party in an economic transaction has more or better information than the other.

- **Consequences:**
 - Market **inefficiency**
 - Potential **market failure**
 - **Types:**
 1. **Hidden Information** (Before the contract)
 2. **Hidden Action** (After the contract)
-

2. Adverse Selection (Hidden Information)

- Arises when **sellers know more** than buyers (or vice versa) **before** the transaction.
- Classic example: **Used car market**
 - Seller knows if a car is a "lemon" or not, buyer doesn't.
 - Leads to **low-quality products dominating the market**.
- **Insurance Market:**
 - Riskier people are more likely to buy insurance → premium increases → healthy people exit the market.

Solutions to Adverse Selection

1. **Restrict Opportunism:**
 - Universal health coverage
 - Anti-fraud laws
 2. **Equalize Information:**
 - **Signaling:** Informed party reveals info (e.g., education as a signal of ability)
 - **Screening:** Uninformed party induces the other to reveal info (e.g., different insurance contracts)
 - **Third-party information:** Ratings, reviews, certifications
-

3. Moral Hazard (Hidden Action)

- Arises when **one party changes behavior** after a contract because it's not being observed.
- Examples:
 - Taking a loan and using it recklessly.
 - Insured driver driving carelessly.

Solutions to Moral Hazard

1. **Efficient Contracts:**
 - Designed to align incentives.
2. **Types of Contracts:**
 - **Fixed Fee:** Pay a set amount regardless of outcome.
 - **Contingent Contracts:**
 - **State-contingent:** Payoff depends on external conditions.
 - **Profit-sharing:** Both parties share profits, incentivizing effort.
 - **Bonuses:** Performance-based incentives.
3. **Monitoring:**
 - Audit, supervision, performance evaluation
 - Can happen **during** or **after** the activity

4. Principal-Agent Problem

- A special case of moral hazard.
- **Principal** (e.g., employer) hires an **agent** (e.g., worker).
- Problem: Agent may act in self-interest, not in principal's interest.
- Need for **incentive-compatible contracts** and **monitoring**.

5. Externalities

- **Definition:** Uncompensated impact of one person's actions on others **not involved** in the transaction.

● Negative Externalities

- Harmful effects on others:
 - Pollution
 - Noise
 - Smoking
- **Markets overproduce** the good → **Social surplus not maximized**

● Positive Externalities

- Beneficial spillovers:
 - Research
 - Immunization
 - Well-maintained property
- **Markets underproduce** the good → Also inefficient

💡 Optimal Solution:

- Move market from private equilibrium to **socially optimal** quantity using:
 - **Taxes** (for negative externalities)
 - **Subsidies** (for positive externalities)
 - **Regulations**
 - **Property rights & Coase Theorem**

◆ WEEK 11 – Asymmetric Information & Externalities

◆ Expected Cost with Adverse Selection

$$E(\text{Cost}) = \sum P_i \cdot \text{Cost}_i$$

◆ Principal-Agent Utility

$$U = \text{Expected payment} - \text{Effort cost}$$

◆ Externality Correction via Tax (Pigouvian Tax)

$$\text{Tax} = \text{Marginal External Cost}$$

◆ Socially Optimal Quantity

- Set $MSC = MSB$, where:
 - $MSC = \text{Marginal Social Cost} = \text{Private Cost} + \text{External Cost}$
 - $MSB = \text{Marginal Social Benefit}$

Week 12

PART 1: Public Goods

Characteristics of Public Goods

1. **Non-Rivalrous:**
 - One person's use doesn't reduce availability for others.
 - Example: Watching fireworks.
 2. **Non-Excludable:**
 - Can't prevent anyone from using it.
 - Example: Lighthouse, streetlights.
-

Four Types of Goods

	Excludable	Non-Excludable
Rivalrous	Private Goods	Common Resources
Non-Rivalrous	Club Goods	Public Goods

Problems in Provision

- **Free Rider Problem:** Individuals benefit without paying.
 - Leads to **under-provision** by private markets.
 - **Government** steps in to:
 - Tax and provide
 - Subsidize provision
-

Solutions

- Government funding and regulation
 - Voluntary provision (inefficient)
 - Club models for partial exclusion
-

PART 2: Platform Economics I

What Is a Platform?

- An **intermediary** that facilitates interaction between groups of users.
 - Examples:
 - **E-commerce:** Amazon, Flipkart
 - **Social media:** Facebook, Twitter
 - **Payment systems:** Visa, PayPal
 - **Exchanges:** Stock market (BSE/NSE)
 - **App stores:** Apple, Google Play
-

Network Effects

- Platforms are defined by **network externalities**.

Positive Network Effects:

- More users = More value.
- Example: WhatsApp becomes better with more users.

Virtuous Circle:

- More buyers → Attracts sellers → Attracts more buyers → Reinforces seller base.

Types of Network Effects

Type	Description
Direct (Same-side)	More users on same side → more (or less) value to each other
Indirect (Cross-side)	More users on one side → more value to the opposite side

Examples:

- **Direct Positive:** Social platforms (WhatsApp users benefit from each other).
- **Indirect Positive:** More gamers → More game developers (e.g., Xbox).

Platform Externality Map

Platform	Group A (→ B)	Group B (→ A)	Within A	Within B
Visa	+	+	-	+
Dating Apps	+	+	-	+
Xbox	+	+	/	/
Newspapers	+	-	/	-

Multi-sided Platform vs Conventional Business

- Traditional: One-side (Seller → Buyer)
- Platform: Two/multi-sided, creating value by managing network effects on **both** ends.

PART 3: Platform Economics II

Refined Network Effects Concepts

➤ Attraction Loop (Direct / Same-Side):

- e.g., Google Maps: More users → More data → Better product → Attracts more users

➤ Attraction Spiral (Mutual Cross-Group):

- More buyers attract sellers; more sellers attract buyers.
- Leads to **"Winner-Takes-All"** dominance.

➤ Attraction-Repulsion Pendulum:

- Cross-group effect is **positive on one side, negative on the other**.
- Example: OS users vs malware creators.

➤ Attraction Spillover:

- One group benefits from another, but reverse doesn't happen.
- Example: Professionals don't care about amateurs, but amateurs benefit from pros.

📈 Modeling Network Effects

Simple Model:

- 2 users: A and B
- Value of product = intrinsic value + network benefit (β)

Results:

Type of Network Effect	Outcome
Weak ($\beta < r_A - r_B$)	Standard demand pattern (can have only one buyer).
Strong ($\beta > r_A - r_B$)	Multiple equilibria possible (either both buy or none).
	Creates unpredictability and inefficiency .

🧠 Key Insight:

- In presence of strong network effects:
 - **Users' decisions become interdependent**
 - **Adoption might fail** even when all would be better off with adoption

◆ WEEK 12 – Public Goods & Platform Economics

◆ Expected Total Utility of Public Good (e.g. fireworks)

$$\text{Total Utility} = \sum U_i(Q)$$

◆ Free Rider Problem Utility

- Users enjoy benefit *without contributing* to cost.

◆ Two-Sided Market Cross-Externality Value

$$V = V_{\text{inherent}} + \beta \cdot N_{\text{other side}}$$

◆ Simple Platform Pricing Rule

- Maximize:

$$\text{Profit} = N_A \cdot P_A + N_B \cdot P_B - C(N_A, N_B)$$

◆ Network Effect Model (2 consumers A, B)

$$r_i = r_{\text{inherent}} + \beta \cdot (\text{\# of adopters besides } i)$$

- If $\beta > r_A - r_B \rightarrow$ multiple equilibria possible